

Nasdaq TotalView-ITCH

Dataset ID: [XNAS.ITCH](#)

Nasdaq disseminates full depth-of-book data for their exchange through their TotalView-ITCH feed over the MoldUDP64 protocol, which Databento receives over UDP multicast in our NY4 data center.

PDF specifications are available on Nasdaq's website for [Nasdaq TotalView-ITCH ↗](#) and [MoldUDP64 ↗](#).

Symbol convention

Raw symbols in this dataset use the **Nasdaq** convention (e.g. `BRK.B`, `AAC+`). See [CMS vs. Nasdaq symbology](#) for the per-dataset convention map.

Timestamps

Nasdaq TotalView-ITCH messages received by Databento have at most one timestamp, which has nanosecond precision. As with all of our datasets, we also collect a `ts_recv` timestamp when the packet was received by our capture server.

The extra field `ts_in_delta` does not convey any additional information for this dataset, and is equivalent to `ts_recv - ts_event`.

Databento Field	TotalView Field	Description
<code>ts_recv</code>	N/A	The capture-server-received timestamp.
<code>ts_event</code>	Timestamp	The matching-engine-received timestamp.

More details about our timestamps are available in our [timestamping guide](#).

MBO normalization

Order lifecycle messages

There are seven message types in TotalView that change the state of the order book. Databento normalizes these messages to MBO records as follows:

- *Add Order - No MPID Attribution*: **Add**
- *Add Order with MPID Attribution*: **Add**, but the attribution field is not included
- *Order Executed*: **Trade**, **Fill**, then **Cancel**
- *Order Executed with Price*: **Trade**, **Fill**, then **Cancel**, but the initial **Trade** is omitted if the message is marked as non-printable
- *Order Cancel*: **Cancel**, with the `quantity` indicating the number of shares to remove from the display size
- *Order Delete*: **Cancel**, with the `quantity` indicating the current display size of the order
- *Order Replace*: **Cancel** with the previous `order_id`, followed by **Add** with the new `order_id`

All Trade records will have `order_id` set to `0`, because Nasdaq does not provide information about the aggressor's order. Normalized records from the same TotalView message will have the same `sequence`.

Trade messages

TotalView also has two messages for providing execution details for match events against non-displayed orders or cross events.

Databento normalizes both *Trade (Non-Cross)* and *Cross Trade* messages into a single MBO Trade record with side **None**.

Nasdaq will always send a *Cross Trade* message following an opening, closing, or EMC cross event for an instrument. If no shares are matched